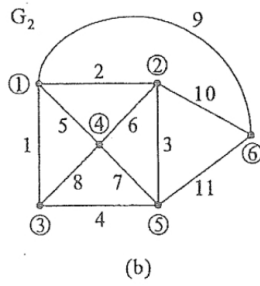
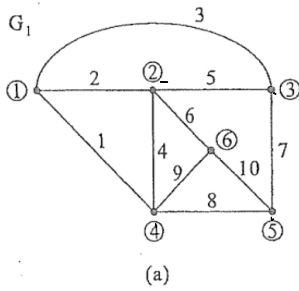
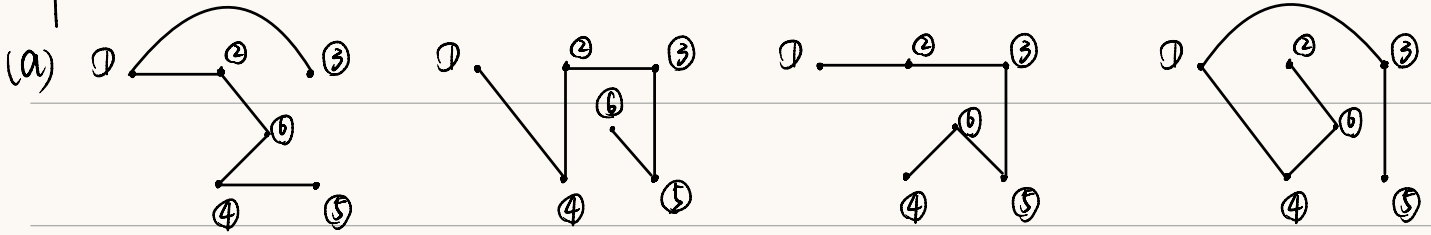


3-3

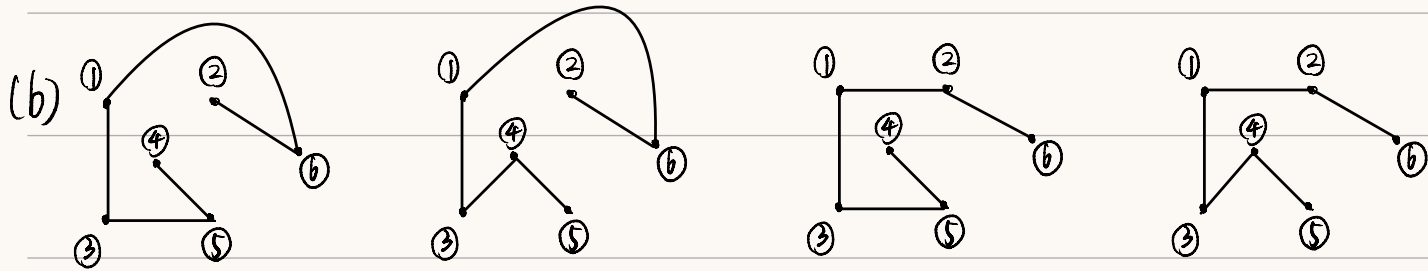
对题 3-3 图(a)(b)所示图,各画出 4 个不同的树,树支数各为多少?



解:



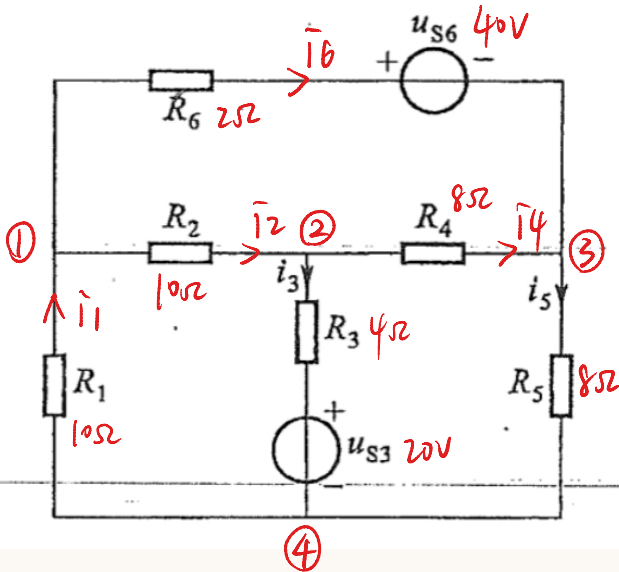
树支数:  $b = n - 1 = 5$



树支数:  $b = n - 1 = 5$

3-7

题 3-7 图所示电路中  $R_1 = R_2 = 10 \Omega$ ,  $R_3 = 4 \Omega$ ,  $R_4 = R_5 = 8 \Omega$ ,  $R_6 = 2 \Omega$ ,  $u_{S3} = 20 \text{ V}$ ,  $u_{S6} = 40 \text{ V}$ , 用支路电流法求解电流  $i_5$ 。



解: 如图, 对 3 个网孔列 KVL

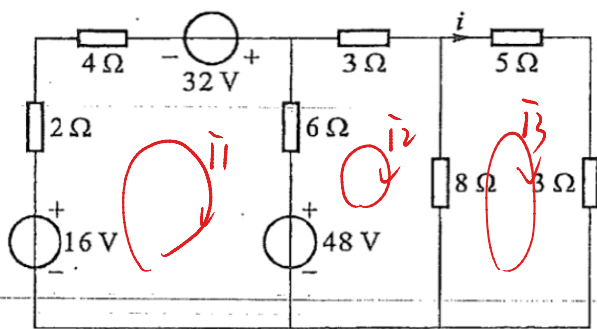
$$\begin{cases} u_{S6} + i_6 R_6 - i_4 R_4 - i_2 R_2 = 0 \\ i_4 R_4 + i_5 R_5 - u_{S3} - i_3 R_3 = 0 \\ i_2 R_2 + i_3 R_3 + u_{S3} + i_1 R_1 = 0 \end{cases}$$

再对 ①、②、③ 列 KCL

$$\begin{cases} i_1 = i_2 + i_6 \\ i_2 = i_3 + i_4 \\ i_4 + i_6 = i_5 \end{cases}$$

代入方程得  $i_5 = -0.956 \text{ A}$

3-10 用回路电流法求解题 3-10 图所示电路中  $5\Omega$  电阻中的电流  $i$ 。



题 3-10 图

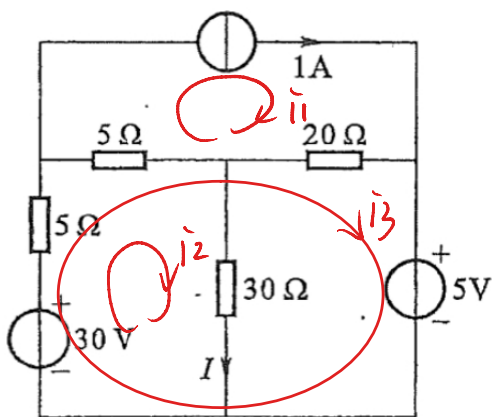
解：对 3 个回路分别列 KCL

$$\begin{cases} (4+2)i_1 + 6(i_1 - i_2) = 32 + 16 - 48 \\ 3i_2 + 8(i_2 - i_3) + 6(i_2 - i_1) = 48 \\ (5+3)i_3 + 8(i_3 - i_2) = 0 \end{cases}$$

$$\Rightarrow \begin{cases} i_1 = 2.4A \\ i_2 = 4.8A \\ i_3 = 2.4A \end{cases}$$

$$\therefore i = i_3 = 2.4A$$

3-11 用回路电流法求解题 3-11 图所示电路中电流  $I$ 。



题 3-11 图

解：对 3 个回路列 KCL 方程

$$\begin{cases} i_1 = 1 \\ 30i_2 + 5(i_2 - i_1) + 5(i_2 + i_3) = 30 \\ (5+20)(i_3 - i_1) + 5(i_3 + i_2) = 30 - 5 \end{cases}$$

$$\Rightarrow \begin{cases} i_1 = 1A \\ i_2 = 0.5A \\ i_3 = 1.5A \end{cases}$$

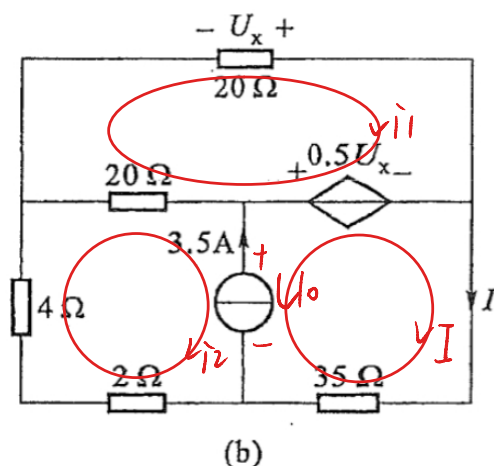
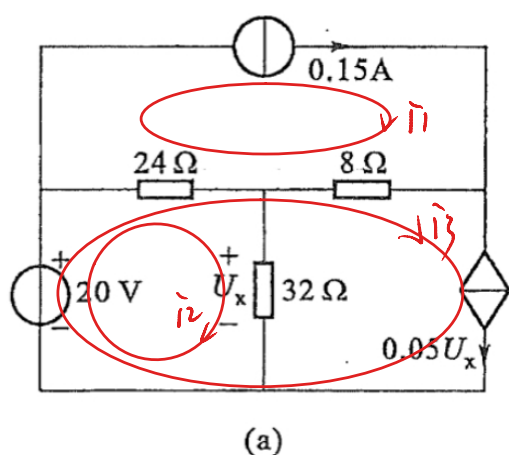
$$\therefore I = i_2 = 0.5A$$

~~3-13~~ 用回路电流法求解：

(1) 题 3-13 图(a)中  $U_x$ 。

(2) 题 3-13 图(b)中  $I$ 。

(在本题中,应考虑如何选择独立回路,可使解题的工作量最少。)



题 3-13 图

解: (1) 对3个回路列KCL方程

$$\begin{cases} i_1 = 0.15 \\ 32i_2 + 24(i_2 + i_3 - i_1) = 20 \\ i_3 = 0.05U_x \end{cases}$$

对  $U_x$ , 有  $U_x = 32i_2$

$$\Rightarrow i_2 = 0.25A$$

$$U_x = 32i_2 = 8V$$

(2) 对3个回路列KCL方程

$$\begin{cases} 20i_1 + 20(i_1 - i_2) = 0.5U_x \\ (4+2)i_2 + 20(i_2 - i_1) = -U_0 \\ 35I = U_0 - 0.5U_x \end{cases}$$

对  $U_x$ , 有  $U_x = -20i_1$

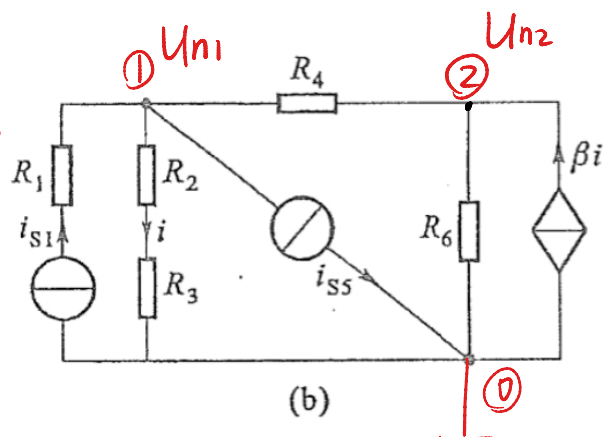
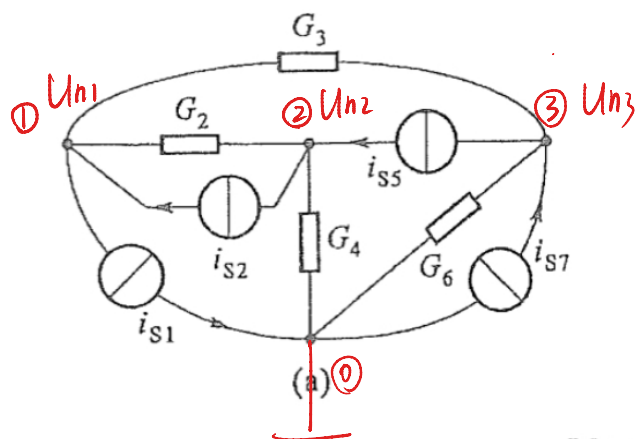
$$\text{且 } I - i_2 = 3.5A$$

$$\Rightarrow i_2 = -2.5A$$

$$\therefore I = i_2 + 3.5A = 1A$$

3-15

列出题 3-15 图(a)(b)所示电路的节点电压方程。



题 3-15 图

(a) 如图, ①结点为参考结点

对①、②、③列KCL有

$$\begin{cases} (G_2 + G_3)U_{n1} - G_2U_{n2} - G_3U_{n3} = i_{S2} - i_{S1} \\ (G_2 + G_4)U_{n2} - G_2U_{n1} = i_{S5} - i_{S2} \\ (G_3 + G_6)U_{n3} - G_3U_{n1} = i_{S7} - i_{S5} \end{cases}$$

(b) 如图, 以①为参考节点

对①、②列KCL有

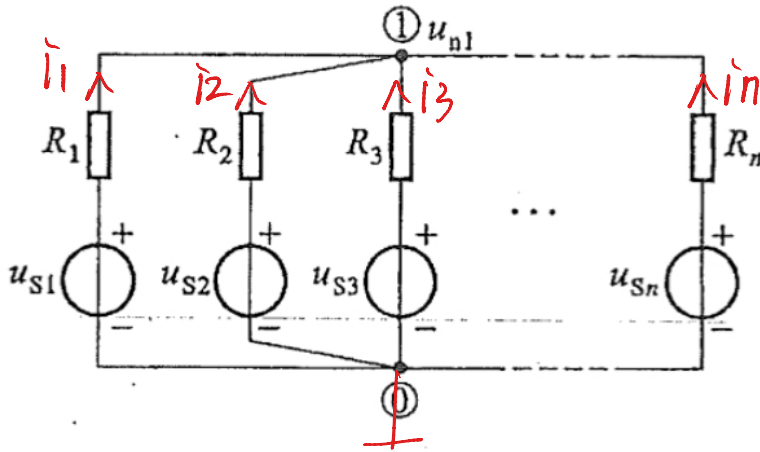
$$\begin{cases} \left( \frac{1}{R_2 + R_3} + \frac{1}{R_4} \right) U_{n1} - \frac{1}{R_4} U_{n2} = i_{S1} - i_{S5} \\ \left( \frac{1}{R_4} + \frac{1}{R_6} \right) U_{n2} - \frac{1}{R_4} U_{n1} = \beta i \end{cases}, \quad i = \frac{U_{n1}}{R_2 + R_3}$$

$$\text{代入得} \begin{cases} \left( \frac{1}{R_2 + R_3} + \frac{1}{R_4} \right) U_{n1} - \frac{1}{R_4} U_{n2} = i_{S1} - i_{S5} \\ \left( \frac{1}{R_4} + \frac{1}{R_6} \right) U_{n2} - \left( \frac{1}{R_4} + \frac{\beta}{R_2 + R_3} \right) U_{n1} = 0 \end{cases}$$

3-17 题 3-17 图所示为由电压源和电阻组成的一个独立节点的电路, 用节点电压法证明其节点电压为

$$u_{n1} = \frac{\sum G_k u_{sk}}{\sum G_k}$$

此式又称为弥尔曼定理。



题 3-17 图

解: 设参考节点为②

对①列 KCL 方程,  $(\frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_n}) u_{n1} = i_1 + i_2 + \dots + i_n$

$$\text{即 } \sum_{k=1}^n G_k \cdot u_{n1} = \sum_{k=1}^n i_k$$

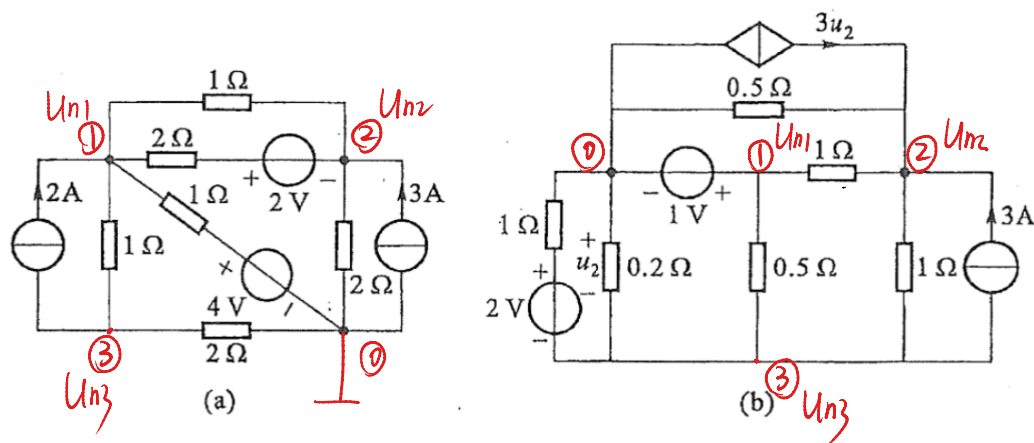
$$\text{其中 } i_k = \frac{u_{sk}}{R_k} = u_{sk} \cdot G_k$$

$$\therefore \sum G_k \cdot u_{n1} = \sum u_{sk} \cdot G_k$$

$$\Rightarrow u_{n1} = \frac{\sum u_{sk} \cdot G_k}{\sum G_k}$$

3-18

列出题 3-18 图(a)(b)所示电路的节点电压方程。



题 3-18 图

(a). 如图所示, ③为参考结点

$$\begin{cases} (1+1+\frac{1}{2}+1)U_{n1} - (1+\frac{1}{2})U_{n2} - 1 \cdot U_{n3} = 2 + \frac{4}{1} + \frac{2}{2} \\ (1+\frac{1}{2}+\frac{1}{2})U_{n2} - (1+\frac{1}{2})U_{n1} = 3 - \frac{2}{2} \\ (1+\frac{1}{2})U_{n3} - 1 \cdot U_{n1} = -2 \end{cases}$$

整理得

$$\begin{cases} 3.5U_{n1} - 1.5U_{n2} - U_{n3} = 7 \\ 2U_{n2} - 1.5U_{n1} = 2 \\ 1.5U_{n3} - U_{n1} = -2 \end{cases}$$

(b). 如图所示, ③为参考结点

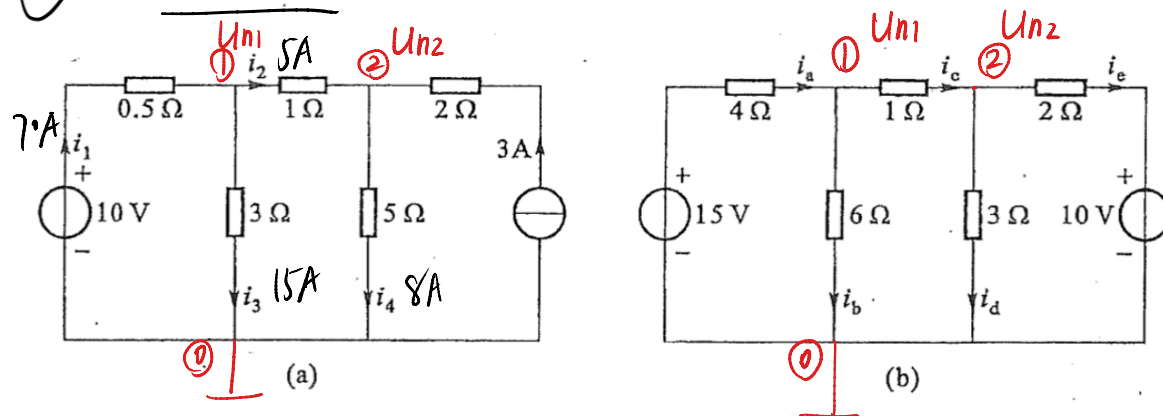
$$\begin{cases} U_{n1} = 1 \\ (1+\frac{1}{0.5}+1)U_{n2} - U_{n1} - U_{n3} = 3 + 3U_2 \\ (1+1+\frac{1}{0.5}+\frac{1}{0.2})U_{n3} - \frac{1}{0.5}U_{n1} - U_{n2} = -3 - \frac{2}{1} \end{cases}$$

控制量  $U_2 = -U_{n3}$ 

代入化简得

$$\begin{cases} 4U_{n2} + 2U_{n3} = 4 \\ 9U_{n3} - U_{n2} = -3 \end{cases}$$

3-19 用节点电压法求解题 3-19 图所示电路中各支路电流。



题 3-19 图

(a) 以①为参考结点, 有

$$\begin{cases} (\frac{1}{0.5} + \frac{1}{3} + 1) U_{n1} - U_{n2} = \frac{10}{0.5} \\ (1 + \frac{1}{5}) U_{n2} - U_{n1} = 3 \end{cases} \Rightarrow \begin{cases} U_{n1} = 9V \\ U_{n2} = 10V \end{cases}$$

$$\therefore i_1 = \frac{10V - U_{n1}}{0.5\Omega} = 2A$$

$$i_2 = \frac{U_{n1} - U_{n2}}{1\Omega} = -1A$$

$$i_3 = \frac{U_{n1}}{3\Omega} = 3A$$

$$i_4 = \frac{U_{n2}}{5\Omega} = 2A$$

(b) 对①、②列 KCL

$$\begin{cases} (1 + \frac{1}{4} + \frac{1}{6}) U_{n1} - U_{n2} = \frac{15}{4} \\ (1 + \frac{1}{3} + \frac{1}{2}) U_{n2} - U_{n1} = \frac{10}{2} \end{cases} \Rightarrow \begin{cases} U_{n1} = \frac{171}{23} V \\ U_{n2} = \frac{156}{23} V \end{cases}$$

$$i_a = \frac{15V - U_{n1}}{4\Omega} = 1.89A$$

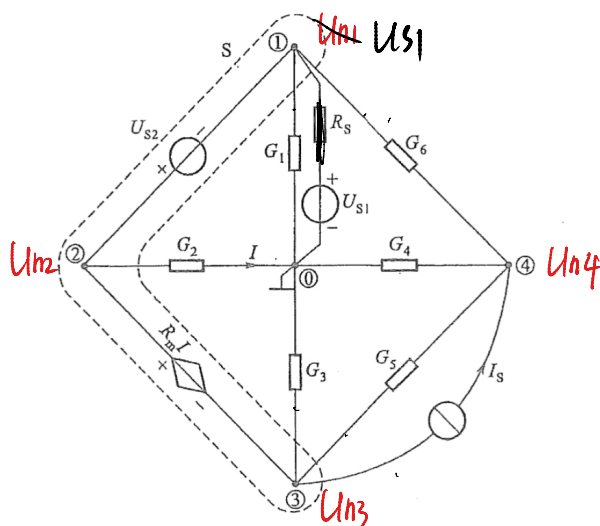
$$i_b = \frac{U_{n1}}{6\Omega} = 1.24A$$

$$i_c = \frac{U_{n1} - U_{n2}}{1\Omega} = 0.65A$$

$$i_d = \frac{U_{n2}}{3\Omega} = 2.26A$$

$$i_e = \frac{U_{n2} - 10V}{2\Omega} = -1.61A$$

3-26 列出题 3-26 图所示电路的节点电压方程。如果  $R_s = 0$ , 则方程又如何?  
 (提示: 为避免引入过多附加电流变量, 对连有无伴电压源的节点部分, 可在包含无伴电压源的封闭面 S 上写出 KCL 方程。)



题 3-26 图

解: 对 (1 2 3) 封闭面和 4 列 KCL 方程

$$\text{有 } \begin{cases} (G_6 + G_4 + G_5) U_{n4} - G_6 U_{n1} - G_5 U_{n3} = I_s \end{cases}$$

$$\begin{cases} (G_1 + G_6 + \frac{1}{R_s}) U_{n1} + G_2 U_{n2} + (G_3 + G_5) U_{n3} - (G_6 + G_5) U_{n4} = \frac{U_{s1}}{R_s} - I_s \end{cases}$$

$$\text{约束方程 } \begin{cases} U_{s2} = U_{n2} - U_{n1} \\ R_m I = U_{n2} - U_{n3} \end{cases}$$

$$\text{受控量 } I = U_{n2} \cdot G_2$$

若  $R_s = 0$ , 则  $U_{n1} = U_{s1}$

$$\begin{cases} (G_6 + G_4 + G_5) U_{n4} - G_6 U_{s1} - G_5 U_{n3} = I_s \end{cases}$$

$$\begin{cases} (G_1 + G_6) U_{s1} + G_2 U_{n2} + (G_3 + G_5) U_{n3} - (G_6 + G_5) U_{n4} = -I_s \end{cases}$$

$$\text{约束方程 } \begin{cases} U_{s2} = U_{n2} - U_{s1} \\ R_m I = U_{n2} - U_{n3} \end{cases}$$

$$\text{受控量 } I = U_{n2} \cdot G_2$$